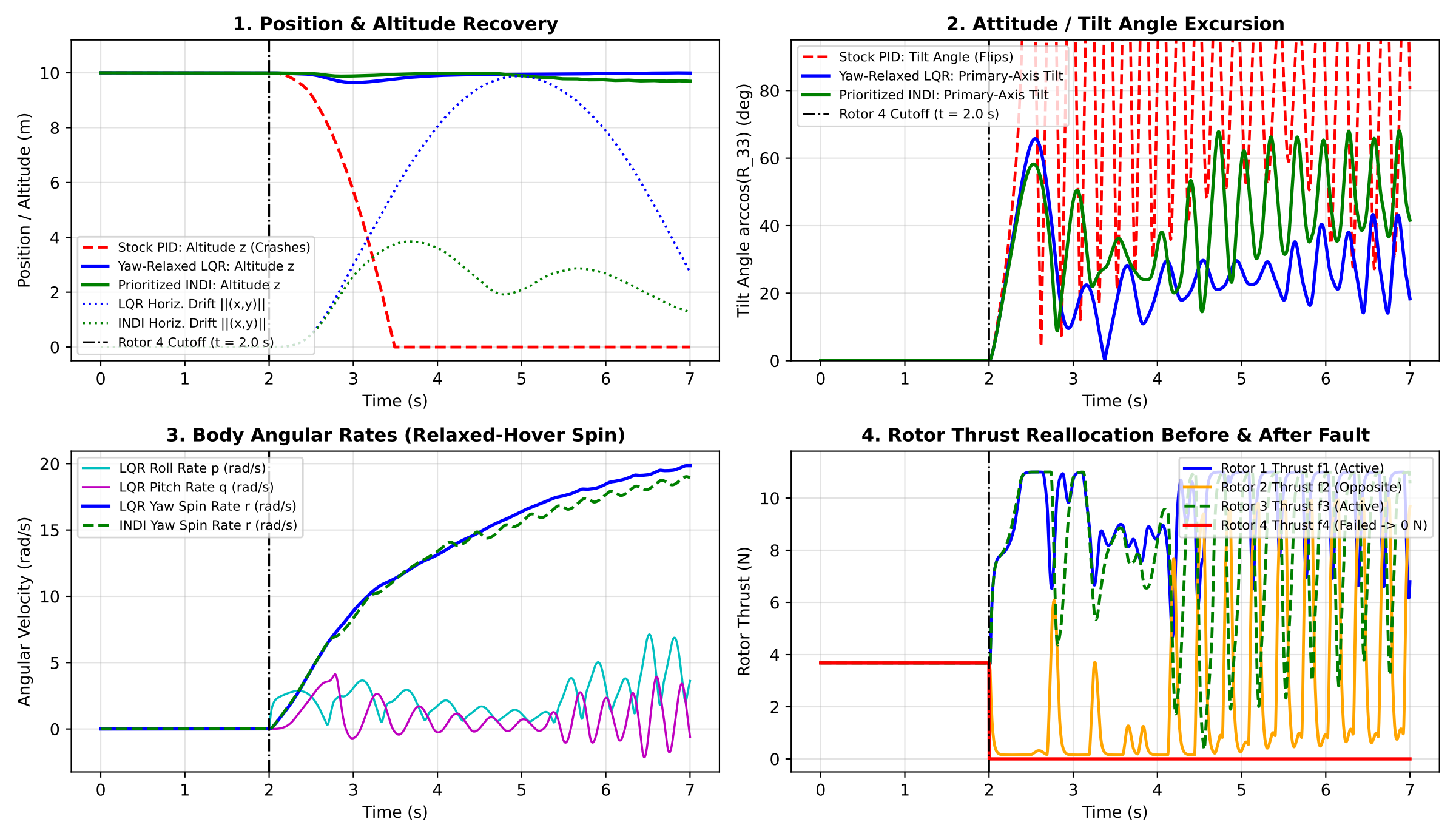

1. Position & Altitude Recovery
Position / Altitude (m)
Time (s)
Stock PID: Altitude z (Crashes)
Yaw-Relaxed LQR: Altitude z
Prioritized INDI: Altitude z
LQR Horiz. Drift ||(x,y)||
INDI Horiz. Drift ||(x,y)||
Rotor 4 Cutoff (t = 2.0 s)
2. Attitude / Tilt Angle Excursion
Tilt Angle arccos(R_33) (deg)
Time (s)
Stock PID: Tilt Angle (Flips)
Yaw-Relaxed LQR: Primary-Axis Tilt
Prioritized INDI: Primary-Axis Tilt
Rotor 4 Cutoff (t = 2.0 s)
3. Body Angular Rates (Relaxed-Hover Spin)
Angular Velocity (rad/s)
Time (s)
LQR Roll Rate p (rad/s)
LQR Pitch Rate q (rad/s)
LQR Yaw Spin Rate r (rad/s)
INDI Yaw Spin Rate r (rad/s)
4. Rotor Thrust Reallocation Before & After Fault
Rotor Thrust (N)
Time (s)
Rotor 1 Thrust f1 (Active)
Rotor 2 Thrust f2 (Opposite)
Rotor 3 Thrust f3 (Active)
Rotor 4 Thrust f4 (Failed -> 0 N)